Agentic Delegation and the Language Frontier

A model, GitHub evidence, and a Lean audit

Quispe & Xu (2026) - arXiv:2605.25438v2

Course replication: ai-03-quispe

github.com/davidolano03/ai-03-quispe

First check: the requested citation was stale

PROMPTED: Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers

VERIFIED v2: Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub

Alexander Quispe & Kevin Xu

Takeaway: title, authors, version date, and sample size were checked first.

Question and contribution

Can an agentic coding tool move developers beyond their prior language frontier?

1. Delegation model of extensive-margin entry
2. GitHub evidence around voluntary Claude Code adoption
3. Predictions for specialists and lower-ability developers
4. Formal audit of all five propositions

The agent's economic role

Before: $1\{V0_l >= 0\}$

With agent: $1\{\max(V0_l, VA_l) >= 0\}$

New activation: $V0_l < 0 <= VA_l$

agent threshold |==== activation band ====| solo threshold

Takeaway: delegation lowers the threshold for attempting an unfamiliar language.

Five propositions, one logical chain

P1 Better of solo/delegated value weakly expands the frontier.

P2 Expansion equals probability mass in the activation band.

P3 Repeated activation creates a cumulative gap with diminishing increments.

P4 Effects rise with unfamiliar candidates and activation probability.

P5 Lower repository-entry costs expand breadth; strictness needs positive mass.

Data and identification

5,346 developers: 2,813 early adopters; 2,533 not-yet-treated controls

149,688 developer-months, Jan. 2024-Apr. 2026

3.2m commits and 57m changed files

First Claude-coauthored commit marks adoption

Callaway-Sant'Anna doubly robust staggered DiD; one-month anticipation

Caution: adoption remains voluntary.

Main event-study magnitudes

Outcome Adoption t+1 t+2

Languages +2.528 +1.227 +0.693

New languages +1.193 +0.126 -0.018

Language entropy +0.382 +0.189 +0.102

Pre-period means: 0.90 languages; 0.31 new languages

A sharp spike leaves persistent breadth; new entry quickly fades.

Is this only Claude writing the code?

Exclude first-Claude language: +1.58 languages; +0.807 new languages

Remove every Claude-coauthored commit: +1.663; +0.724

Roughly two-thirds of diversification remains

Per-commit diversification concentrates at adoption

Persistence partly reflects higher activity volume

Takeaway: broader than mechanical counting, but not randomized.

Where I distrusted the paper: Proposition 3

$G(t) = 1 - (1-p2)^{(t+1)}$ when $p1 = 0$

Delta $G(t) = p2(1-p2)^{(t+1)}$

Paper permits $p2 = 1$; then $G(t) = 1$ for every t

Strict growth and strict concavity therefore fail at the endpoint

Strictness also fails when the unfamiliar-language set is empty

VERDICT: require a nonempty set and $0 < p2 < 1$.

AUTHENTIC HAND-DERIVATION PHOTO

Add hand/derivation.jpg before submission

From the paper to Lean

Original: G(t) = sum_l [(1-p1_l)^(t+1) - (1-p2_l)^(t+1)]

Lean proof uses Finset.sum_lt_sum plus an explicit witness language.

hNonempty supplies the witness; hClosedFrontier supplies $p1=0$ and $0<p2<1$.

All five exact-type proof endpoints compile.

No sorry, admit, new axiom, opaque, or native_decide.

Assumptions made explicit

1. Foothold: augmentation requires prior familiarity; delegation drives entry.

2. Verification technology: comparative statics require signed cost effects.

3. Comparable candidates: unfamiliar languages share an activation probability.

Lean checks the implication structure, not empirical truth of assumptions.

Comparison: two ways to model AI

QUISPE-XU

AI = information, delegation, verification

Margin = domain/language entry; risk = project selection

AOUAD-LYKOURIS-ZHONG

AI = substitutable productive input

Margin = task allocation/human capital; risk = deskilling

Modeling choice determines which economic mechanism becomes central.

Three critiques

1. The entry threshold may capture a generic productivity shock, not uniquely delegation.

2. Voluntary adoption cannot separate causal expansion from project selection.

3. Strict propositions need endpoint conditions: positive mass, nonempty sets, and interior hazards.

The descriptive pattern is strong; mechanism and causality are less settled.

Conclusion

Agentic AI appears to broaden what developers attempt.

Large, partly persistent expansion in language breadth

Coherent extensive-margin delegation mechanism

Robustness checks reduce but do not eliminate selection concerns

Formalization validates weak results and repairs the strict dynamic claim

github.com/davidolano03/ai-03-quispe